Collective Behavior (Discussion) ~ Daniel Friedman ~ Active Inference for the Social Sciences 2023

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hello and welcome it's August 16 2023 we're in the collective Behavior discussion section of the active inference for the social sciences course in 2023. so perhaps let us all go around and give a opening introduction and or reflection and see who joins and what happens from there but just so that everybody has said hello Jim hi everyone I'm Brynn and uh 9 pm in South Africa and uh yeah just excited to be here see what happens hi I'm Andrew I'm a graduate student in Indiana and um I guess my field of uh interest now is the Knowledge Management and I think that figuring out how we collectively learn and can share information my teacher is a really interesting topic and I'd like to see how this would apply Regina and I'm performing with them now that I'm living in Spain studying and doing experiments related to this topic and trying to learn a little bit of the um Theory that's behind it so it can you know Advance better I'm excited hello my name is Sasha I'm calling in from California I'm also a graduate student um yeah just really enjoyed the spirited presentation last week I'm looking forward to a discussion all right well how about anyone can give a first reflection on Collective Behavior or what what was their thought or context before and where are they at as we kind of begin yeah Regina okay so I just want to make sure that I understand because I took like two big lessons from your lesson right and the first one is that um you know this this famous model of active inference that has the agent and the environment and the Markov blanket in the middle right and just interacts with everything right so I think what I got from what you discussed um maybe I didn't have a good definition of an agent but first that you can have that agent not be just an individual but also your definition of an individual changes depending on what you want to study in the behavior right so an agent can be an aunt it can be an ecosystem it can be an eye looking at something or it can be a as a society for example and I think that's a big take for me because it changes a lot the concept of how to study things and use active imprints just by changing how big the agent can be so I don't know if like I got that right right so first that clarification so what is an agent yeah so I don't know if there's like a formal definition of an agent in active inference right but uh if we if we want to study whatever we want to study because um I'll study because I have to do that's like my goal whatever I'm doing so I want to study whatever I have to choose what's going to be my agent and the environment it has so choosing the agent you can do it to create a model or to make an experiment whatever but um the choice you make it will depend on your in what you understand an individual or as an agent I think so right awesome anyone have a thought on that no but if if I continued you know just to add something else and this will be my my um and then I'm thinking is that also a big problem that we have or I think I have is that since I am part of the environment um you know you know the perspective of whoever is studying whatever it is it's also going to be an effect in it so maybe maybe scientists already decided but I don't know that we can never have an objective way of studying something if you're part of the environment of the agent so I don't know we already forgot it or we already decided that's it we cannot have objective observations of behavior in general but you know you always think about like the agent environment and you studying it as something else but this something else is also effect in it so I see that as a problem or maybe something that has to be said like we are not objective because we're part of what it's affecting our ages in the behavior collectively too so those are the the thoughts I have to share uh Regina I'm uh I'm also new to to all of this so my preliminary understanding of of orders and agent is uh anything that has any good reason uh to self-evidence and to itself or to survive um that doesn't want to dissipate that might delineate the boundary of what an agent is that then sits at uh um the boundaries with which it um will as you described start interacting with the world with the environment so if something as I understand it if something has a reason to go on being to um then it is an agent in the world okay but but that agent could be anything right like is that what you're saying right like if he wants to survive it can be an ant surviving individually or a cell or or a group okay yeah that's a good that's like a good definition okay yeah Sasha um great points Regina um so much to say there um something that I have experienced with um as a researcher in biology is deciding the kind of the statistics or sampling approach to use whether we're taking you know one cell from one individual or we're taking many cells from many individuals and how to do the statistics and kind of um sampling approach on that um that is a you know an evergreen question that we're always reevaluating to see um how to think about the independencies within biological systems and then another thing that you mentioned was about the individual being part of the ecosystem and that makes me think a lot about complexity science and something that um is addressed there in uh thinking about the interplay of the scientists and what they're studying yeah thank you for those points especially I can sorry I'm not raising my hand I'm just I'm using it that's okay can you say a bit more about what you mean by complexity designs is baby muted oh can't speak at the moment yeah complexity science has looked at Asian asia-based modeling as a fundamental tool John Holland signals and boundaries focusing on what from an active inference side we might say it's like message passing or communication for the messages and then boundaries with the whole discussion around the boundaries in the world map territory map territory fallacy map territory fallacy fallacy doing iterative modeling Sasha brought up limited experimental resources in in the scientific setting and that's kind of analogous to The Limited energy budget or attentional budget ultimately or just decision-making budget of uh the kinds of active entities that make sense to model the ones that kind of have those uh characteristics this is really the question of what makes something first principles or what are the first principles one aspect of it as brought up is the censoring of the persistence so from the outside that's kind of like an empiricists view of persistence like is it measurable but not saying that something is necessarily that which is the statistical tool and then the sort of like inside out implementation type but there's a lot more to say in that who wants to bring in some New Angle or how do they see Collective Behavior why do we talk about Collective Behavior why not just Behavior recognize that behavior is multi-scale or or social what's really being communicated with the collective I mean I I believe that we chose those levels so we can make it more simple for our brains to understand it because if we talk about Collective Behavior actually is the universe behaving and the particles and everything right so I don't know if that's a little bit just based on I don't know what's the word like just in the material world and how it physically moves and that's like deterministic or whatever right I think us we chose to have those limitations so we could understand the world but also we should acknowledge that are limited because we're part of the system I'm having a little trouble on my kind of understanding the I guess the where the Schism lies between kind of the the Markov blanket and uh the um well in uh semiotics of like the animals uh sensory experience anyone have a maybe maybe someone could uh I mean Daniel Dino like the difference between that too um can you be a little louder thank you though Andrew anyone have a thought on that I'll bring in a definition of whom felt then and then we can see but meanwhile does anyone have a thought sorry sorry I was just muted on Andrew you're a little quiet but um I'm gonna bring in a definition of umvelt and then meanwhile does anyone have a thought I I hope I'm not uh where the diluting Andrew's uh uh question I think a collective behavior in kind of like a I don't know football team or soccer team um and what that provides uh the grip who is uh share some common ethos and what that does to the how that regulates the their Collective or individual and Collective free energies um during a match or throughout the season and how that that gives some yeah some charge to uh some Investments because it's my team against yours and we have to um try and win and so you know without it without that Collective Behavior it's it's might be far harder to achieve the group's goals if they aren't able to collaborate and share um you know implicitly and explicitly certain traits behaviors thoughts feelings ideals ideologies whatever they might be and how that's been allows us to compete in the world I I thought you think it's yes okay okay I see what you're saying uh especially because we're watching over the World Cup right Women's World Cup yes it's very relevant yeah but I see that in all the there's things you've said it seems that for your behavior needs to have like a goal and I don't know if I if I agree with that it's either survival or winning but main for me it's just something that happens and maybe the consequences surviving but it's not the goal to survive or to win the game I mean in in the football it is right but in general um maybe there's no goal to the behavior collectively and just just a thought it's interesting yeah all the great comments well the paper of Connor Hines at all on the collective Behavior as surprise minimization that points the way towards using the actin fontology these kinds of modeling approaches for Collective behavioral situations and really just recognizing them as a certain type of Behavioral modeling the simplest case might just be a single closed system all part of the composition that leads to the ability to describe at very least a lot of different kinds of systems just like a linear model can describe a lot of systems so then it's also really interesting a topic that came up in one of the recent textbook groups with what's normative about active inference because the term of uh the process Theory even of active inference being normative is mentioned multiple times and that's definitely a topic that connects to the social sciences one can imagine the kind of Niche normativity or niche regularities in a broad sense and then the real minute particulars of how they play out which kind of hints at the kind of descriptive or causal minimum to start making some anything but toy specific generative models so it's been an interesting theme to see even already in avells and Ben's lectures and discussions this question between description process of describing which is what I associated with Professor Gordon's approach to ant watching how that descriptive process which is what we can say that we have for active inference just like a linear modeling ecosystem could say about their linear modeling in principle um and how to then articulate the normativities of the framework if that concept if that separation can even exist from the uh Behavioral normativities that include the basis of social Fabric in all different kinds of communities here's another question so what is the role of any technical detail in in the social sciences whether more on the statistical connecting with data or more category theoretic-like just formal what's the role of formal methods in different kinds of academic or practical Expressions about social systems doesn't matter at all how we think about a framework at all well for formalizing aspects of a social system that I would say that the purpose is for communication between individuals because just you know we can't verify that well formalizing it is just setting a system on top of what you're observing um but there is no explicit uh guarantee that you're saying encompasses everything of the phenomena you're observing and just like you know kind of going back to taoism of like the name that can be named is not the true Eternal name and so it's just a way of you know it's it's a slice of what we can observe the way to um you know maybe you can make predictions or maybe you can generate something that acts like the the phenomena phenomena you're observing but um so it's just it's a way to kind of it's just a way to kind of approximate our understanding of some phenomena and you have to slice it in a specific way but also understanding that it is a slice or you know it is a tangent line that you're approximating as opposed to encompassing the whole system from verbalizing it um I will uh maybe it's not the goal but something that happens out of describing any system like Andrew said it's like um the goal to modify it or to or to corrupt it in some way to gain control or whatever we know the model you know how to play the game kinda so you write instructions and try to see if you can outsmart someone else I mean it sounds kind of dark right but maybe it's one of the goals to or something that just happens out of knowing the rules that nobody else knows I mean I guess when you when you talk about control there is a there there is a shade that can be um gained from that specific wording but control in terms of you know control theory or you know that that comes from electrical engineering it's really just what what shapes the behavior or what are they what are the inputs that drive the the behavior and when you're talking about control it's really just about um well if you change these inputs how does that change the end behavior and so you can actually you know from electrical engineering of controlling a some sort of electromechanical system but then you can make analogous statements on like control into society which doesn't necessarily mean like the dark negative control but like you know maybe there's some pathological behavior in in a society or uh you know even pathological pathological responses within the brain and it's like well what is what is driving that and is it possible to to change those those change the drivers of that to have something that is a more beneficial to the system yeah just

just to add um to what Andrew's saying because it matters sometimes how you manipulate um uh you know a a system or it matters how you affect a brain so it cannot be random I mean uh you know we've come a long way for example from lobotomy uh which had disastrous effects and and you you it matters to a system how you affect it and so to know more formally uh the most effective way to create a desired effect whatever that might be and it's essential that it's not just kind of poking and prodding there's like that that has [Music] as much as that has is part of the scientific method to update that method in accordance with what you're learning so it is is important no I think we're both writing it doesn't have to be dark but also something else that Andrew said about predicting I think that's what also the the big goals trying to predict whether to be prepared to or or just to I mean it couldn't be to control right but if you if you can somehow predict how people respond to days or how a system will behave in this other scenario you can you know change what the input is or prepare for whatever comes I guess a lot of great points some of the commonly held not not equally by everyone everyone but some of the attendance of science being to explain predict design and control where explanation and prediction are on the pathway towards uh systems designing and and maintenance and other other words that people can add but that's the pragmatic utility oriented View which doesn't have to be an absolutist view but certainly there are regimes of attention socially now and seemingly for long long times where people were wanting to retain and to constrain the variance distribution on what we would say is their generative model some Norm and then other times engaging in these functions and that instrumentalist usage of active inference is like we're going to be able to model it with a pretty minimum framework that's commonly used in statistical systems where some things we observe and some we don't and at least we know and how all the pieces are connected and then that is an artifact in model-based design as people develop that line through time socially just like probably has been always the case for a complex projects and scientific collaboration there's a few other interesting pieces yeah please please I'm just I'm just wondering if well now that I'm thinking out loud um isn't that like a problem or confirmation bias right like there's a paper of in here here because we create the knowledge and we tell everyone the knowledge this is the model this is how everything behaves and then we update our knowledge about it and then we look for everything to just you know confirm it and and yeah maybe there's another system that will not let us be getting knowledge so slowly because if everybody just agrees on something we teach it to everyone and we confirm the model and then we just try to look for evidence for that model then we go slowly if there's any better thing or better model that comes along you have a lot to say about that like if we believe that trees exist or that people exist and then we come to see them so there's kind of this sense where problem confirmation bias which I think is a great term a great topic to bring with definitely also what's called like the Golden Nail syndrome or something if all you have is the hammer then everything's a nail and there's definitely attention there because it's like how much of what you can do are you supposed to do something that you can't do or but then at least you could do what you needed to do to get there so that's actually within your scope of what you can do so it's not that there are things that you can't do that you can't do if just from a measurement first membership so like how does it look or how do we recognize across systems or how do we have awareness in situations as limited individuals who do have all of these cognitive phenomena how to make decisions or sense making or infrastructure or whatever it is the social Ness the the system of interest of the social sciences make that and then dot dot dot normatively what I think it's the sun protection built into into uh the problem of confirmation bias because all knowledge generation is happening in in a in an open system and so there's always going to be an agent outside of your your own mind and it's and what it's perceiving that that can then challenge or add to the you know like the the the super the the LK 99 semiconductors that have there's been all the rage recently and and how um at the scientists have now kind of um discredited it uh quite convincingly and it's because of this open they might have had degrees of confirmation bias that they were egging each other on but it's it's ultimately been refuted which is uh this built-in kind of protection because it's all operating in a in a in an open system fortunately yeah a lot to see there uh Monday morning reviewer touring how how do we again go from any given description of a communicative event like the kind of specific unfolding of the LK 99 narrative as yet uncompleted in in the minds of sum which is real enough to be real socially and uh so how do we go from any descriptive anything about that total observation to core screening then how do we come to uh work with that interface with what whatever that is hands that to us what is the social behavior what are some some classic or some novel social behaviors maybe ones that seem pretty obvious due to recency bias I'll just mention one aspect that Andrew and I have also talked about is the auto poetic nature of Information Systems Andrew do you have any thought on this or about where Auto places comes into like social sciences or these kinds of models they're genuine Andrew just left a note that he's uh okay okay we just wasn't looking at the time funny stuff anyone else have a thought on that okay just just to kind of carry it forward then basically the the ability of social systems that include written Niche modification cumulative culture they're increasing extent of control as discussed earlier that that definitely is a bit of a overloaded or at least doubly meant term in different settings because it can just simply mean modeling it with control theory which is pretty limited philosophical statements relative to any statistical method or control like the kind that people are unhappy with and the kind that's described in thinking like a state and a lot of other works that I know Mao and Lorena and aval are going to be definitely going into more uh detail on so when societies can model and introspect or or self-reflect formally much better and that can happen through anything from free and open source Federated protocols there's many limited formats of self-meaning making but cryptography definitely allows those Explorations and that doesn't necessarily mean computer cryptography that could also be like linguistic encoding and all these other ways of of niche sub differentiation in multi-scale systems you just did not see a system that is cookie cutter it's very rare in ecosystems there's algal blooms and things like so when we really think about the Developmental and the cognitive and the selective niche which is a concept that Axel constant and others have differentiated in some of their like ecological octave inference work like that Niche is very embodied extended and cultured etc etc it's just like a kind of cool direct way open up to that challenge of modeling bring some closures with different accounts or compatibilities with different accounts so like all just to give one random example but maybe it'd be interesting what what other people think all the Expressions that people make in Social settings can just simply be their expressions or modeled as such that is in one way implicitly what was always in play but now it's describable and legible and autonomous so how do we take this active inference Collective Behavior crossover and think about what it really means in how we're going to be like thinking differently even just this discussion forward or like more locally are our understandings with Collective Behavior or with behavior Daniel can you ask the the question again I I didn't actually catch the um how do we the quiz start how do we start to adapt some of these General comments or or frames that are in some ways disconnected from descriptions of real systems how do we think about that and and what it means now what is on the yeah good that's okay uh do you mean on the document in front of us or these these these disconnected things in front of us that on on the on the document I don't know whatever you think or whatever you want to say yeah I I think there's a lot uh and I again I hope I'm not you know and fully understanding the question I hope I'm not steering away uh too much but I gotta had a general sense of what I think you were saying and I think it's an enormous uh problem with the amount of um information sharing um just in a in a with let's just take uh the general uh generative models of um social media and all of the platforms that are available to us how much information we are privy to in any given uh moments I'm kind of forgetting the question that I that you asked um but but how do you it's almost as if there is such a a high rate of information and data flow that it's really hard to know what to do with all of it and and at least as a human being I'm not saying for machines but um but uh as a psychologist I'm certainly interested in how humans are able to tolerate and respond to the amount and rate of information that's all okay this is not the first time this happens right like we went from talking to writing information then everyone wrote their information and then we have books and then we have libraries and that's all available and again we have access if you can read right to all the information in the world so we adapted to I don't know to get careers so we could specify in which type of information you are going to know more so maybe we need some sort of new career types I don't know more specifications with more information not uh is the rate of information flowing increases and not is it not exponential is it not uh um I know it's it's a crazy amount that our brains cannot do anything about it so we have to do multi-disciplinary things and work together with others if we want to make something yeah I don't know maybe we will die of excess of information how did people feel like active inference bring something to the discussion relative to how they want to work towards ants yeah well isn't that the provocative question too but like any system that's that that's the question everyone is gonna have their own memes and themes and and systems that they have spent time with and did field work with and all this other like particularity because a system that doesn't have particularity is just not going to be hosting a broad enough Niche for for phenotypic diversity uh it's just riding on the edge of a razor instead of more like a fabric that continues to unfold I think there's a lot of work to do with the social sciences and active inference just from what I was like reading trying to find things for this um discussion and just I personally wondered what is is Meaningful or interesting or useful what's epistemic and pragmatic value for difference audiences How does it go from kind of bringing citations that were not linked together some more theoretical or more generative model oriented work in active inference like epistemic foraging or the infra ants model or any other kind of specific model what are these different kinds of scientific artifacts do for who I guess I'm still learning very very kind of crude be cuddling along here Daniel with to try and answer that because it's I've quickly realized that the field is it's so it has so much um the the free energy principle uh the Bayesian brain even just to psychology has so much applicability and everyone's seen everyone I meet at these discussions is coming at it from a different angle um and it can be a bit confusing to hear how other people uh are applying it to how other people are um making use of it um in all manner of ways and so I'm still learning how to apply um to kind of ring fence the free energy principle and connect it to uh psychological processes including Collective behavior and because there's a lot of Psychopathology in group think in in Collective Behavior but I'm also very very interested in its application at the interface of um what happens between a a therapist and a patient and and I'm often unlike other um professions who seem who are just by nature of the scientific method um are able to extract a lot more data to work with to analyze to um uh um you know um in a that's very hard to do I feel very limited um other than just making kind of theoretical leaps and inferences about how to apply um uh the free energy principle to psychology uh it's I can't have a session with my brain hooked up to an fmri machine to get data and to see uh when a patient says that's a spike on the the spike trains which represents um you know surprisal and we can then solve for that in some way I'm I don't have access to that sort of data flow so it feels limited but I'm still learning how to apply it yeah and trying to collaborate with colleagues on that the influence group what are people wondering about even if we're totally of course not going to address it but just what what are the kinds of things that people what makes you look up Collective Behavior instead of just Behavior or individual foreign safety and need to belong for me it's um I think that um it depends for where you come from and the country you are but there are places that are more Collective and there are places that are more individualistic and I think mostly non-western countries have a lot of collective Behavior or knowledge and periods already right that they already knew these things that we think are new for us right how the Collective thing works for example I don't know you don't make a decision for yourself you make a decision for the group like marrying you marry because you want an audition you want to interact with someone or other group and that's something that affects the the collective not just you so arrange marriage makes sense something like that right so for me making that decision about individual Collective comes from understanding that I don't know there are other places that are already ahead in what Collective means in terms of human beings and how to interact and the Western Wall we have all these individualistic thing which is very interesting because why do we have the same species and then we have very different approaches of how to understand ourselves in terms of what an individual is and why it matters yes there's so much loaded on different understandings among people including even the the essence of the which is who are people and what is the the obligation and how do we view that that relationship that's a very very deep area and and again the part that just kept returning to me in the lecture and when I was trying to think about how to how to even reflect part of uh this very large area is like what is the adjective doing and what what is what's the reference point by which this is being said in the speech acts like people often will dismiss I don't mean to um you know straw people but dismiss Collective Behavior by saying well we are explaining individual Behavior what are you explaining Beyond The Nest mates in The Colony and then the exterior like that is it and so that can feel like to looking at the same thing from two different sides and there are different methods like some methods blur over individuals a lot more like statistical methods or that might correspond to demographics and then on the other hand like more agent-based but how much variability in what shared variability and why General synchranies amongst different Quantum reference frames all these nested if these possibilities are are grounded in wanting to explore mathematical and scientific possibilities as as the literature of linear regression is full of so many incredible extensions and and and similar and then that's I I think what the course is even beginning to to try to express and what I understand in the works of the other teachers like a lot of your backgrounds and and everything like that is where is that moment where it is not only science because science is not simply the social so science has an activity and then as there's a diffusion away from science and I think that or or at least that relationship between social and science is right there in the title maybe that's kind of like a science Majors take but it but it is itself in a way a um presenting a dilemma with what science means there and then that's the frame game who's doing social science who's doing not social science and that's that's where um in in the culture here I'm sitting in slash in the technical side of that it is related to statistics and formality it's the xkcd comic and the way that formalisms and and details are taken as under some regimes of attention basically definitive and yet there's like a yearning and a counterbalancing that's expressed also in many ways and so that's and science maybe some people feel like certain social scientific methods that's like not far enough in this Dimension and then the others feel like they're too far in this direction but it's like just like any uh like that academic field let alone any other community of diverse agents like of course there's going to be distributions of perspectives at different times by those different perspectives and you know Daniel could do you think uh these first principles would be able to explain why an ant colony exists uh I know that was your um kind of area of interest of study uh why an ant colony exists uh in in the way that it is executive operating as a collective and why um another um animal is able to survive and achieve its goals as an individual uh only relying on the environment and its actions and Sensations to get its needs met into no doubt minimize its um free energy but could first principles explain why the one has to survive as a group and the other has to survive or is able to survive um under different conditions yeah that's a super interesting question it's it's very core to this individual individuation and multi-scale individuation concept like is the nest made the individual and then the colony is the social because then for sure it's super collaborative or is the colony of the organism and massmates are tissues and then are they social across Colony boundaries not where I was in Arizona but they're often friendly across different species and they avoided conflict more often than not so it's like what and then go why does it even matter what's happening on this plot of lands in this in this area from like a biomimicry perspective we might learn new algorithms it's a Sandbox for learning new techniques and observation all these other like richnesses so it's to say that like what comes away from that science that makes it social science and I think there's a lot of disciplinary ways of understanding that which which I'm definitely not expert in in the um professional Fields like like counseling um but probably in in all the different Humanities and histories and all these other fields all these different Maps the kind of social the epistemic organizing question and then the social organizing question yeah and and the way like Technologies also in that tangle and yeah someone pointed out like this is not a new problem and uh but then it's like well what parts are new and then what's our model by which we're saying that it's like new or too individualist or too Collective on what scale what if they're like really collectivists at this scale but not at this scale or another an example of an ambiguous or at least like um polysomis phenotype of colonies is in some colonies there's partial reproductive activation of like non-reproductive Mass mates and that has often been modeled from like a position of like zero is the law and how are they cheating above this but it's also been approached from a colony level strategic bed hedging so without super deep knowledge of really some of the specifics of the regularities that that Colony algorithm had to persist for so long and to have gotten swept off the table if it was even it's like that's that's a very integrated problem so it helps you approach Collective systems and and learn about them but it's like there's a lot to say about humidity loss in foraging and humidity loss isn't even the whole picture and human foraging is a lot more sophisticated from just like a perception cognition influence perspective so those are just some ant related thoughts but it just centers a lot of discussions that really matter and I think the really interesting piece is not always in the way that they're thought to by some yeah and it's and it's very hard to get any kind of cross-pollination with OTC caused pollination studying One agent it's it's quite for at least in my mind very difficult to kind of generalize things across different agents um it starts getting a bit yeah tricky to infer one collector's uh what how and why and how they survive and how another agent does It's Tricky so I I'm almost human beings as hard as they are to study and just try and run with it as far as possible otherwise it's just information overload body is a complex place too and digital technology virtuality these are such um richly developed topics in the social sciences I only know the tiniest amount from but I think that through participation and inclusion in collaboration and in conversations and everything there's going to be a lot of interesting connections and paths it's being approached seemingly a multiple layers like within the the two layers that just we can focus most narrowly on is like within the realm of science what is science which is kind of like playing this other um role with inertia in some people's minds and in systems what is science who gets to decide what social science is I don't know I'm not in social science but it's system of interest to society and certainly I'm a part of that um so it's like what is the what is good technically what is social scientific and then what is that social scientific how does that gonna interface into the the metaphysical questions and and the normative ethical and uh um decision making with trade-offs and the real reflexive complex decisions that matter also while recognizing this increasingly nuanced way of thinking about different social intelligences what's any other random thought or curiosity about Collective Behavior one point that I would have added into the lecture was just that in distributed systems whether as modeled or as such all behavior is arising from interactions local to individuals even if it's sent through some Wireless mechanism it's like all about the local interactions and so that is a a an arising type process that gives the modeler like one hands-off layer at least if not much more another nice feature about them um they are in a way ing to be composed because they intrinsically contain like at least two layers of consideration not whether there's kind of like a forcing function of one layer and just sort of dangling a bottom layer or whether there's a lot of what we would call emergence or discussive in in that way like that more a rising function can be modeled but in either way then you can kind of wrap that two-layer system and then compose them so that's like what enables in certain situations like parallelization or at least you know that's obviously a big area but that's what allows multi-agent simulation Frameworks to do well on certain very challenging sense making in action selection problems not by guarantee or default because any given slicing and dicing of any given prompting of information may not help you adapt but there are paths that do and don't help persist or not and that's this kind of persistence starting place that we began with Sasha um okay I really resonate with what you mentioned about the local interactions that um even if information is transmitted um over long distances um then the agents behave in a way that is locally relevant and integrates that information and it actually makes me think a lot about the like psychology um therapy angle that um while you uh try to get information about uh the person's like um sort of local environment and behavior or like perhaps you know demographics and age that um they're informational environment like a wealth of what kind of um information they consume might also be really telling about their psychological and like Behavior Uh in interactions or pathologies and then to bring it kind of Full Circle back to um that the researcher is part of the ecosystem that they are particip studying yeah I I think it's just um such a difference in perspective to see that as like a positive and a gift that um the way that you would design and run an experiment today is not the way that you would do it two years from now with more experience and so um that is a a a benefit and like a bonus in the system and not um like a negative perspective of of bias that it's detracting from some underlying uh truth or reality so just a few thoughts oh very interesting anyone wanna add something I think we didn't fully resolve this early question about the umbelt um term from ethology behavioral research observation science using the term to describe the world as it's experienced by a particular organism I think a few different pieces and there's a lot of discussion and discourse so just this is just even like a super short overview the world we might connect to World modeling and there's a lot of interesting work in all kinds of different like pros focus on qualitative systems as well as some artificial or synthetic intelligence World modeling as it experienced as it is experienced is one of the sticky and deep ones because this is the area of phenomenology and so this is like a kind of semi-eternal discussion point so whether whether active inference delivers on world modeling is kind of empirical and just plain whereas as it is experienced is going to be a philosophical question and then by a particular organism that specificity we get by actually making the generative model even if we're just speculating about it it's like talking about experience without an experiencer would be like talk talking about um some agent's Behavior without reference to its niche and then where there is like that complex negotiation I think the humanities and a lot of the social sciences have identified a lot of the structural like top-down and also the personal bottom up motifs of of things being blurred and made complex and so I think part of the social and the scientific as coming only from the scientific side um is like to to meet in the third space and or at least make it work to have a mutual compromise where it's like certain social science social oriented claims based upon this scientific measure that is going to be a judgment call but understanding that conversation where is it a mutual compromise in like what dimensions not simply along the kind of like rigor accessibility or implications access there's just very many things in play in these iterated relationships in ecosystems having a researcher as part of the ecosystem well what is it here's here's the forager question you see the nest made on the ground it's a forger but then we step back it's like okay well it's foraging and then that's the allocation of the forager then its foraging is still part of this sense making this generative sense making with like process understanding even of a Trace which is something that Bruno LaTour and others have have explored with like following a trace in a social setting and using that to to trace Networks so in our digital ecosystems though our information ecosystems is not research happening and so what what does that really mean and then what are those social normativities there's going to be like a meeting or a handoff with with where that really is happening or maybe it's a knocking on the door that's ignores it could be whatever metaphor in that setting it actually is there's no General metaphor because I'd be like oh what's the general metaphor on this experiment which experiment just the general action that we plan to take oh hard to say okay well in our last like connections I guess if anybody types a question live chat we'll look at it otherwise what are any other thoughts or like what what can we learn more about about Collective or what will it mean for our sense making decision making if we understand Collective Behavior better what else are people excited about for the rest of the course like based upon the sections that we've done so far and and how how we can even even start to approach this in in an acknowledged Limited finite way yeah Sasha then than anyone else just added a question to the very bottom of the notes um and it makes me think about um how working on this document collaboratively throughout the last hour um we're able to uh kind of leave a stigmergic trace of things we've discussed and what sort of traits or practices should individual humans I guess I should specify humans should we embody in order to improve cohesion um in Collective Behavior it makes me think of certain psychology traits or uh descriptions of like openness um or conscientiousness um what will make individual humans better participants in yeah I I think to to add to that to just emphasize the importance again only with um human agents um specifically if you have the capacity to think and reflect and the only ones that have that those uh really Advanced capabilities that we know of as yet but to really kind of Leverage in order to make the collective better it is a responsibility on of the individual despite being part of a collective and on in many different on many different fronts to be able to leverage and utilize their own internal States and what they have learned and to be able to think in a to retain the individual the individual's mind and think about the collective I think is so important it's the collective is to become healthier essentially to your point and to improve its yeah to become more conscientious to become more responsible or whatever that might be so to always try and maintain a thinking self is crucial unless you can very easily be usurped into the collective as it's if you just go on any social media media platform you will see individuals rarely thinking from themselves and seemingly kind of just fueling each other and that that causes massive problems so an individual mind is strikes me as being just the most important thing in the future going forward with individuation and collectivization as these like ongoing negotiated compromises that can be also viewed many ways however however they cognitive science coming into play maybe uh historical processes philosophies were used in many settings for even approaching these kinds of social questions and now with computer science and technology it even like is a discussion more about the capacities of measurement and action not that those are new in and of themselves like but thinking like a state like the development of the apparatus the development of the social social systems that can be run on on Open Source Hardware open source software things like that I I don't know if something's generalized across all such groups understanding what environments at least initially things aren't beginning at is just a realist starting point probably no guarantee since most things slash all things are swept off the table in their specifics anyway and how to deal with that kind of like skin on the line type of decision making is very interesting how it's pursued in the in the Sciences because often like many runs are presented or that kind of like like catastrophic failure mode of models is is not considered just like part of the training starting naive so just the way the different social considerations are RR converged upon it's never it's not like reduced to one feature of the social because the social is such an open space as a scientist always impressed by how that writing could kind of open up the um situations which are like mundane but also of course Very deep and meaningful also though mundane any other thoughts or otherwise uh on on Collective Behavior or on the social sciences course or anything otherwise people can have any lost words if they want yeah awesome thank you all for participating David want to add anything yeah well I look forward to uh more questions being asked on the course website and everybody's um engagement with the course is very much appreciated so keep up the collective my scale learning and it's it's a fun course really happy that we could have done this session and very happy to have concluded the session all right thank you bye bye everyone ciao